

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

INFR10074 COMPUTER COMMUNICATIONS AND NETWORKS

Thursday 21st December 2017

14:30 to 16:30

INSTRUCTIONS TO CANDIDATES

Answer any TWO of the three questions. If more than two questions are answered, only QUESTION 1 and QUESTION 2 will be marked.

All questions carry equal weight.

CALCULATORS MAY NOT BE USED IN THIS EXAMINATION

Year 3 Courses

Convener: C. Stirling

External Examiners: S.Rogers, A. Donaldson, S. Kalvala

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. (a) The Internet protocol stack has five layers. From an end-host's point of view, there are many protocols in the application, transport, link and physical layers whereas there is only one protocol, IP on the network layer. What is the reason behind this? In other words, what principle led to such a structure? [2 marks]
- (b) This question relates to Domain Name Service (DNS).
- i. A startup named "Firth of Forth" need to register a domain name, and run a web server. Suppose that the domain name and IP address for the web server is `www.firthofforth.com` and `129.215.33.176`, respectively. Assume that the domain name and IP address of the company's authoritative DNS server is `dns.firthofforth.com` and `129.215.33.247`. Write DNS record(s) as a format of (name, value, type) that need to be inserted into `.com` TLD server and into the authoritative DNS server. [5 marks]
 - ii. Briefly discuss a drawback of frequently changing the IP address associated with a domain name of a service. [2 marks]
 - iii. Briefly discuss how DNS can be used to distribute loads to multiple web servers of a service. [2 marks]
- (c) It is sufficient to check destination port number for UDP demultiplexing whereas TCP demultiplexing needs to check 4-tuple (i.e., source IP address, source port, destination IP address, and destination port). Briefly explain why UDP and TCP need different demultiplexing mechanisms. [2 marks]

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- (d) This question relates to two reliable data transfer protocols, Go-Back-N (GBN) and Selective Repeat (SR). Assume sliding window parameter $N=10$ for each of the two protocols. Assume each ack is 40 Bytes and each packet is 100 Bytes. Note that a sequence number is a byte stream number of the first byte in a segment and an ack number refers to the sequence number of the next byte expected. The sender transfers a file of size 1 MByte (i.e., 10^6 Bytes) to the receiver.

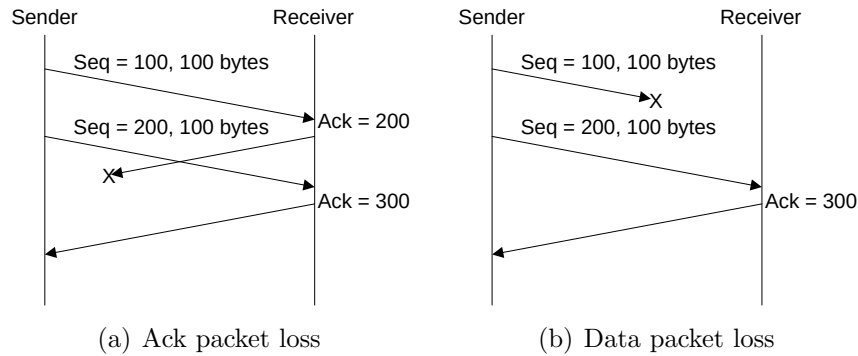
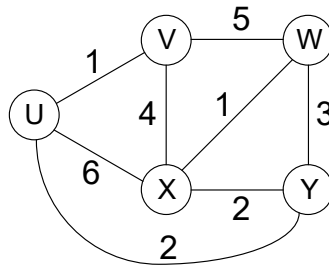


Figure 1: Different loss scenarios.

- i. Consider the scenario in Figure 1(a) where an ack from the receiver is lost. Assume this happens 10% of the time. Compare the two protocols by calculating the average number of packet transmissions in the forward direction (from sender to receiver). [6 marks]
- ii. Consider the scenario in Figure 1(b). In this scenario, suppose that a packet is lost in the forward direction with 1% probability. Compare the two protocols by calculating the average number of packet transmissions in the forward direction. For the GBN protocol, assume the worst case scenario where the first packet of each window gets lost. [6 marks]

2. (a) Consider a simple network where a router connects a pair of hosts. All the interfaces of the router and two hosts have the same bandwidth of 8 Gbps, and the RTT between the two hosts is 100 ms. Assume that there is no packet corruption in the network.
- In this network, congestion never happens. Briefly explain why. [2 marks]
 - When there is only one connection between the two hosts (i.e., one sender transfers data to one receiver), what is an optimal window size in bytes for a pipelined reliable data transfer protocol to fully utilize the network bandwidth? Justify your answer briefly. [3 marks]
- (b) Briefly explain Head-of-the-Line (HOL) blocking that can occur in forwarding packets in the router. Comment on how this problem can be avoided or resolved. [4 marks]
- (c) Consider the following network consisting of 5 nodes.



Using Dijkstra's algorithm, find out the shortest paths to all nodes from node u . Show your work in the form of a table following the template below. Let $D(v)$ be the current value of cost of path from source to destination v ; $p(v)$ be the predecessor node along the path from source to v ; and N' be a set of nodes whose least cost path is definitively known. To get you started, the first two columns in Step 0 are given. [5 marks]

Step	N'	$D(v), p(v)$	$D(w), p(w)$	$D(x), p(x)$	$D(y), p(y)$
0	u	1,u			
1					
2					
3					
4					

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(d) This question relates to Software-Defined Networking (SDN).

- i. Briefly explain three key characteristics of SDN. [3 marks]
- ii. A software-defined network is built with OpenFlow switches. In the network, host X generates a number of TCP flows to host Y. Each flow is distinguished by the typical 5-tuple (i.e., source and destination IP addresses, source and destination ports, and protocol). Suppose packets belonging to all of the flows traverse the same path consisting of switches *A*, *B*, *C* and *D* (hence, the path is *A-B-C-D*). Now consider a problem where packets from one flow never arrive at host Y while those of all other flows do. An operator suspects that one of the four switches drops the packets of the problematic flow, of which the 5-tuple is $\langle X, Y, P1, P2, 17 \rangle$ where X and Y denote source and destination IP addresses, P1 and P2 are source and destination ports, and 17 is the protocol number for UDP. Assume that the flow has an infinite number of packets; thus the operator can perform diagnosis while the flow is alive. Using OpenFlow operations, devise and discuss a methodology or procedure to diagnose which switch is responsible for this reachability issue. Explain any drawback(s) of your methodology. [8 marks]

3. (a) This question relates to multiple access protocols.
- i. Briefly explain four desired properties of an ideal multiple access protocol. [4 marks]
 - ii. The efficiency of the pure (unslotted) ALOHA protocol is twice as low as that of the slotted ALOHA protocol. Briefly explain why. [2 marks]
- (b) Two CSMA/CD stations are each trying to transmit long (multiframe) files. After each frame is sent, they contend for the channel, using the binary exponential backoff algorithm.
- i. What is the probability of success after the m -th collision? [2 marks]
 - ii. Provided that the first collision occurred, what is the probability that the m -th collision can occur ($m \geq 2$)? [4 marks]
- (c) Consider the 4-bit generator, $G = 1001$, and suppose that the data portion of the bit stream to be transmitted prior to Cyclic Redundancy Check (CRC) calculation is 11001001. Show the actual bit stream transmitted. Suppose the leftmost bit in the transmitted bit stream is inverted due to noise on the transmission link. Show that this error is detected at the receiver's end. [4 marks]
- (d) Consider the following notation to model client-side buffering for video streaming. Let B denote the size of the client's application buffer in bits. Let $Q(< B)$ denote the number of bits that must be buffered before the client application begins playout. Let r denote the video consumption rate during playback. Finally, assume that the client receives data into the application buffer from the network at a rate x (referred to as the *fill rate*) when the client buffer is not full.
- i. When $x < r$, determine the length of each continuous playout and freezing period as a function of Q , r and x . [6 marks]
 - ii. If $x > r$, then at what time, say t_f , does the client application buffer become full? [3 marks]